Local Semantic Feature Aggregation-Based Transformer for Hyperspectral Image Classification

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***Abstract*— Hyperspectral images (HSIs) contain abundant information in the spatial and spectral domains, allowing for a precise characterization of categories of materials. Convolutional neural networks (CNNs) have achieved great success in HSI classification, owing to their excellent ability in local contextual modeling. However, CNNs suffer from fixed filter weights and deep convolutional layers, which lead to a limited receptive field and high computational burden. The recent vision transformer (ViT) models long-range dependencies with a self-attention mechanism and has been an alternative backbone to CNNs traditionally used in HSI classification. However, such transformer-based architectures designate all the input pixels of the receptive field as feature tokens in terms of feature embedding and self-attention, which inevitably limits the ability for learning multiscale features and increases the computational cost. To overcome this issue, we propose a local semantic feature aggregation-based transformer (LSFAT) architecture which allows transformers to represent long-range dependencies of multiscale features more efficiently. We introduce the concept of the homogeneous region into the transformer by considering a pixel aggregation strategy and further propose neighborhood-aggregation-based embedding (NAE) and attention (NAA) modules, which are able to adaptively form multiscale features and capture locally spatial semantics among them in a hierarchical transformer architecture. A reusable classification token is included together with the feature tokens in the attention calculation. In the last stage, a fully connected layer is used to perform classification on the reusable token after transformer encoding. We verify the effectiveness of the NAE and NAA modules compared with the traditional ViT through extensive experiments. Our results demonstrate the excellent classification performance of the proposed method in comparison to other state-of-the-art approaches on several public HSIs.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) classification, multiscale, neighborhood-aggregation-based attention (NAA), neighborhood-aggregation-based embedding (NAE), transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are represented by hundreds of narrow spectral bands in the electromagnetic spectrum, including abundant spectral information and discriminative characteristics. This entails great potential to distinguish materials precisely and addresses a wide range of applications in many scientific fields, e.g., feature registration [1], target detection [2], [3], and mineral mapping [4], [5]. HSI classification focuses on distinguishing and assigning exclusive labels to pixelwise objects by identifying spectral and spatial characteristics and has been an attractive topic in the remote sensing community [6], [7], [8], [9]. However, the high dimensionality and the cost of manual labeling, which leads to the Hughes phenomenon and the limited availability of reference data, pose great challenges for HSI classification.

Over the past decade, many works have shown that including spatial information can effectively improve the classification performance. A large number of spectral–spatial feature extraction-based approaches have been developed using the information of neighboring pixels. Textural and structural features are extracted using certain filters and rules, and then the features are fed to the classifier. Among them, Gabor filters [10], [11], local binary patterns [12], [13], and edge preserving filters [14], [15] have been adopted to increase HSI classification performance. Besides, spatial contextual methods have been broadly studied in HSI classification. Many works aim to improve the existing models by adopting local contextual information in the task of discriminating pixels, including sparse representation (SR) [16], [17], [18] and collaborative-representation-based [19], [20] classification methods, and support vector machines (SVMs) [21], [22]. For instance, Luo *et al.* [23] integrated multistructure properties into a unified embedding structure to achieve the complementarity of multiple informations. Duan *et al.* [24] discovered the multivariate sparse relationship between manifold neighborhoods by designing a geodesic-based neighborhood SR model. These methods revealed the discriminant characteristic of HSI and achieved satisfying performance. However, a specific filter is usual required to manually extract the desired spatial contextual features from the HSIs, and the performance heavily relies on careful parameter settings.

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In turn, the deep-learning-based methods are capable of automatically discovering potential relationships and valuable representations in massive amounts of data. Inspired by the success of deep learning backbone networks in other fields, many advanced networks have been widely used in HSI classification, e.g., autoencoders (AEs) [25], [26], [27], convolutional neural networks (CNNs) [28], [29], [30], graph convolutional networks (GCNs) [31], and inception networks. Within these deep learning backbone networks, CNNs have drawn significant attention and achieved remarkable success in HSI classification. Hong *et al.* [32] considered the performance bottleneck of a single model and investigated the performance of multiple tasks to combine the advantage of CNNs and GCNs by exploring multiple fusion strategies. Roy *et al.* [33] used the hybrid CNN architecture to represent more abstract spectral-spatial features from HSI but without reducing the computation costs compared with the traditional 3-D-CNN-based model. Sun *et al.* [28] introduced the attention module into a spectral–spatial convolutional network to suppress the interfering pixels, which weaken the discriminative ability and robustness of the model. Based on the spectral and spatial attention modules, Zhu *et al.* [29] embedded a residual block into a sequential spectral–spatial feature learning network, avoiding overfitting and yielding competitive classification performance. Li *et al.* [30] introduced a two-stream CNN backbone network to boost the contextual information learning ability of models by extracting local and global spatial features from HSIs, and further proposed a specific regularization and normalization strategy to adaptively balance the weights of the aforementioned parallel branches.

Recently, transformers have achieved great success in natural language processing and other fields. Among them, Dosovitskiy *et al.* [34] introduced the vision transformer (ViT) for image classification without convolutional filters. This indicates that transformers can perform outstandingly in image prediction and provide an alternative architecture to the CNN backbone. More recently, the transformer backbone networks have been used for HSI classification, owing to their excellent global context modeling ability. Hong *et al.* [8] considered the spectral sequence of neighboring bands and designed a pure transformer-based SpectralFormer (SF) backbone network, representing sequence attributes of spectral signatures. In addition, the SF achieved competitive performance in comparison to other backbone networks. Sun *et al.* [35] captured high-level semantic features by introducing a Gaussian weighted token module into the transformer architecture, yielding promising performance in terms of classification accuracy and computational complexity. Based on the spatial attention and spectral association modules, Zhong *et al.* [36] proposed a spectral–spatial transformer architecture and designed a factorized architecture search framework to determine the layer-level operation choices and block-level orders.

Although transformer-based architectures have already proven their powerful ability for modeling long-range dependencies and achieved encouraging performance in HSI classification, previous studies usually designated all the inputs as token features in the embedding and attention modules, which limits the ability of the transformer in learning multiscale features and introduces a significant computational burden. To overcome this issue, we propose a local semantic feature aggregation-based transformer (LSFAT) architecture for HSI classification, which allows the transformer to represent the spatial contextual information of multiscale features more efficiently. As illustrated in Fig. 1, the proposed method consists of a simple CNN feature extraction module and the proposed neighborhood aggregation (NA)-based transformer architecture. First, we design a 3-D-CNN-based feature extraction block based on principal component analysis (PCA) and a 3-D convolutional layer to extract shallow features from HSI. Then, inspired by the typically low spatial resolution and high spectral resolution of HSIs, we introduce the concept of homogeneous region into the transformer and develop pixelwise neighborhood aggregation-based embedding (NAE) and neighborhood aggregation-based attention (NAA) modules, which adaptively merge pixelwise token features within the homogeneous region to improve the representation of large objects. In our transformer architecture, the reusable classification token is included together with the token feature before NAA calculation, and then separated after training. By doing this, the classification token can learn long-range dependencies of multiscale features, and a fully connected layer is then adopted to achieve the final classification. The main contributions of the proposed method can be summarized as follows.

1) We propose two innovative NAE and NAA modules that offer a neighborhoodwise representation of multiscale features for HSI classification while significantly improving the classification performance and reducing the computational cost.
2) Based on the aforementioned NAE and NAA, we propose an LSFAT architecture, which is able to preserve pixelwise token features for fine-grained information and also adaptively merge the token features for large-scale structures.
3) We investigate the effectiveness of NAE and NAA in comparison to ViT. The experimental results demonstrate that our LSFAT architecture outperforms several state-of-the-art approaches in terms of classification performance on three public datasets.

The remainder of this article can be summarized as follows. Section II summarizes previous work on the CNN backbone networks and self-attention mechanisms for HSI classification. Section III introduces our proposed method, which comprises a simple CNN-based spectral–spatial feature extraction block and our LSFAT-based transformer architecture. Section IV illustrates the parameter settings and experimental results with respect to other state-of-the-art approaches, and it also provides some discussions. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

A. CNN Backbone Networks for HSI Classification

In recent years, many CNNs have been applied to the HSI classification tasks and achieved excellent results. For instance, Chen *et al.* [37] used a 3-D-CNN model with combined regularization to capture effective spectral–spatial

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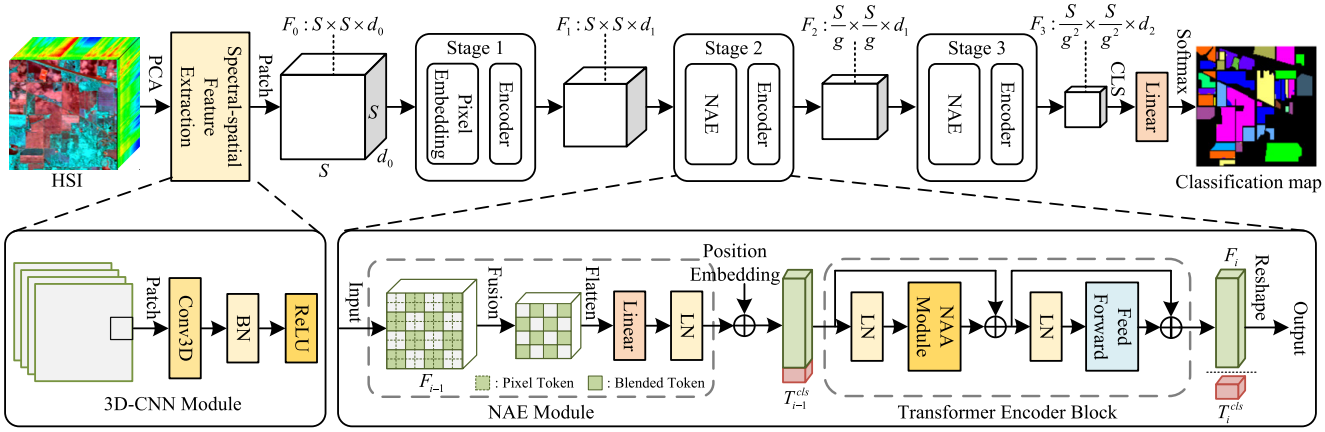


Fig. 1. Schematic view of the LSFAT architecture. The model consists of a 3-D-CNN module and a hierarchical transformer architecture. Each stage of our transformer architecture contains a NAE (or pixel embedding) layer, a NAA layer, and a feed-forward layer. In our transformer architecture, the reusable classification (CLS) token (i.e., T_i^{cls}) is cascaded with the embedded feature (i.e., F_{i-1}) before attention calculation in Stage i.

features from HSIs, which achieved better classification results than the conventional methods. Yang *et al.* [38] designed deeper CNN models to exploit both spatial contextual and spectral-correlation-based features to enhance HSI classification. Paoletti *et al.* [39] improved the CNN model and proposed a spatial–spectral capsule module, providing competitive performance in terms of both classification accuracy and computational time. Besides, Mao designed a novel gated recurrent unit, which reduces the total number of parameters efficiently and yields competitive performance in the HSI classification tasks. Deeper stacked convolutional layers have also been adopted in the CNN backbone networks for HSI classification and achieved promising results. Zhong *et al.* [40] introduced a spectral and spatial residual block and extended the scope of the CNN backbone for extracting discriminative features from spectral signatures and spatial context. Although CNNs have been able to supply inspiring performances in HSI classification, they are usually restricted by a limited receptive field. Besides, the CNN backbone suffers from the high computational cost when attempting to model long-range dependencies.

B. Self-Attention and Transformer for HSI Classification

The self-attention mechanism was proposed and introduced into the CNN backbone to alleviate these drawbacks. Zhang *et al.* [41] designed a spectral–spatial self-attention classification framework integrating long-range dependencies with local features and yielding promising classification performance. Xu *et al.* [42] analyzed the impact of adversarial attacks on the HSI classification task and proposed a self-attention contextual mechanism to improve the robustness of deep neural networks. Moreover, Li *et al.* [43] extended the scope of the self-attention mechanism and defined a novel homogeneity attention block to represent the feature correlation with the homogeneous mask. He *et al.* [44] proposed a bidirectional encoder representation from the transformer module, which effectively captured long-range dependences among pixels. Recently, transformer architectures [8], [35], [36] are used for HSI classification. Specifically, SF [8] captured the sequential properties in spectral dimension using a pure transformer architecture. Different from them, this article introduces a hierarchical transformer architecture for flexible learning multiscale feature.

III. Proposed Approach

An overview of the proposed method is illustrated in Fig. 1. As the figure shows, the proposed method consists of two main components, i.e., a shallow 3-D-CNN module and the hierarchical LSFAT architecture.

A. CNN-Based Shallow Feature Extraction

Let assume that an HSI is denoted as $\boldsymbol{I} \in \mathbb{R}^{M \times N \times D}$, where M and N are the number of samples and lines, respectively, and D is the number of spectral dimensions. Several hundred spectral bands contain abundant discriminative characteristics. However, the high dimensionality also suffers from the Hughes phenomenon, which weakens the prediction ability of models and potentially yields misclassification. Therefore, PCA [45] is used on the HSI to reduce the spectral dimensionality. The modified image is represented as $\boldsymbol{I}_{\text{pca}} \in \mathbb{R}^{M \times N \times d}$, where d is the reduced spectral dimension. Then, the modified image is divided into 3-D overlapping feature patches, the labels of which are decided by the central pixel of each patch.

The CNN module of this work comprises a 3-D convolution layer, a batch normal (BN) layer, and a rectified linear unit (ReLU) activation function. In the process of 3-D convolution [33], the input patch is convolved with a set of 3-D convolutional kernels. The activation value $v_{i,j}^{x,y,z}$ at spatial position (x, y, z) in the jth feature map of the ith layer is generated as follows:

$$v_{i,j}^{x,y,z} = f\left(b_{i,j} + \sum_{\tau=1}^{d_{l-1}} \sum_{\lambda=-\eta}^{\eta} \sum_{\rho=-\gamma}^{\gamma} \sum_{\sigma=-\delta}^{\delta} \omega_{i,j,\tau}^{\sigma,\rho,\lambda} \times v_{i-1,\tau}^{x+\sigma,y+\rho,z+\lambda}\right) \tag{1}$$

where f is the activation function, $b_{i,j}$ is the bias parameters, d_{l-1} is the number of feature map in the $(l-1)$th layer, $2\eta+1$ is the depth of the kernel along the spectral dimension, $2\gamma+1$ is the width of the kernel, $2\delta+1$ is the height of the kernel, and

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$\omega_{i,j}$ is the value of the weight parameter. The parameters of the 3-D-CNN (i.e., $b_{i,j}$ and $\omega_{i,j}$) are usually trained using supervised approaches with the help of a gradient descent optimization technique.

B. Neighborhood Aggregation-based Embedding (NAE)

Inspired by the success of the variable transformer, we extended the applications of shrinking transformer into pixelwise classification and propose a pixelwise NAE module. In HSIs, adjacent pixels within a homogeneous region normally convey similar textural information, which allows us to represent them by a more informative super-pixel. Based on this assumption, this work takes local pixels' aggregation operation for merging the pixelwise token features and learning multiscale features. Pixel embedding and NAE modules are adopted progressively in our transformer architecture for preserving the pixelwise fine-grained information and learning the large-scale structural information.

1) In Stage 1 of our transformer architecture, the pixel embedding layer is first used to preserve the fine-grained information. An input feature F_0 of size $S \times S \times d_0$, extracted by the 3-D-CNN convolution, is fed into the our transformer architecture. Then, the input is reshaped to the pixelwise token features of size $S^2 \times d_0$ and fed to the linear projection layer. The output feature is reshaped and denoted as F_1, with size $S \times S \times d_1$, where d_1 is the embedding dimension of Stage 1. Then, a reusable classification token T_0^{cls} of size $1 \times d_1$ is included in the token feature F_1 as an extra token. Next, the token features (along with a position embedding) are passed through the proposed transformer encoder. After the transformer encoder finishes, the output is divided into the reusable token T_1^{cls} and token feature F_1 for training the next stage.
2) In the remainder Stage i (i.e., Stages 2 and 3), the proposed NAE layer is undertaken to flexibly form multiscale features. F_{i-1} of size $(S/g^{i-2}) \times (S/g^{i-2}) \times d_{i-1}$ is fed into our transformer block as the input feature. A tiny region of HSI is regarded as a homogeneous region, in which adjacent pixels exhibit similar textural information. Therefore, the input F_{i-1} is evenly divided into (S^2/g^{2i-2}) homogeneous regions of size $g \times g \times d_{i-1}$. Then, the pixelwise features of each homogeneous region are blended by a fusion operation to form the blended feature F_i of size $(S^2/g^{2i-2}) \times d_{i-1}$. The token feature F_i is fed to the linear projection layer, and the shape of the embedded feature can be viewed as $(S^2/g^{2i-2}) \times d_i$, where d_i is the ith embedding dimension. An additional fully connected layer is used to adjust the channels of the classification token T_{i-1}^{cls} to match the token feature. After that, the token feature F_i is cascaded with the reusable classification token T_{i-1}^{cls}. The concatenated feature along with the position embedding are fed to the transformer encoder. In a similar fashion, the token features F_i and T_i^{cls} (separated from the extracted feature) are fed as the input to the next stage. Finally, we obtain a classification token T_3^{cls}, and a fully connected layer with softmax is used to conduct classification.

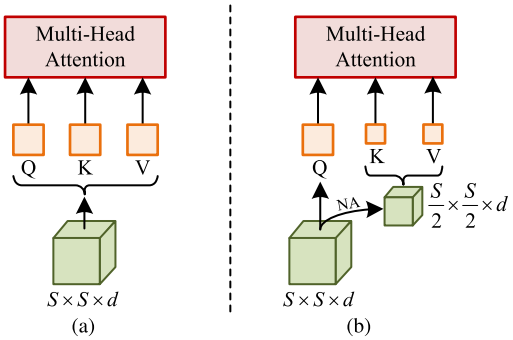


Fig. 2. Comparison of the attention modules. (a) MHA mechanism: the input is projected into Q, K, and V directly in ViT [34]. (b) NAA module: the reduced input is projected into K and V to reduce the computational cost after NA in the proposed LSFAT method.

C. Neighborhood Aggregation-based Attention (NAA)

The transformer encoder block of ViT consists of an attention layer and a feed-forward layer [34]. Fig. 2 illustrates the comparison of the multihead attention (MHA) and NAA modules. As Fig. 2(a) shows, the input is projected into a query Q, a key K, and a value V in the attention layer. Then, the MHA mechanism is adopted to compute self-attention. In our NAA module [see Fig. 2(b)], we perform an NA operation to reduce the scale of the input, so that smaller K and V are obtained, which can reduce the computational burden of MHA. The details of the NAA module are given below

$$\text{NAA}(\text{Q}, \text{K}, \text{V}) = \text{Concat}\left(\text{head}_1, \ldots, \text{head}_{h_i}\right)\text{W}_p \tag{2}$$

where $\text{Concat}(\cdot)$ represents the concatenation operation [46]. W_p are the linear projection parameters of size $d_i \times d_i$. h_i denotes the number of heads, and head_j is defined as follows:

$$\text{head}_j = \text{Attention}(\text{QW}_j^{\text{Q}}, \text{SR(K)W}_j^{\text{K}}, \text{SR(V)W}_j^{\text{V}}) \tag{3}$$

where W_j^{Q}, W_j^{K}, and W_j^{V}, with size $d_i \times d_{\text{head}}$, denote the weighted metrics of Q, K, and V, respectively. d_{head} is the dimension of heads, which is calculated by (d_i/h_i). $\text{SR}(\cdot)$ is the blend operation for reducing spatial sizes, implemented as follows:

$$\text{SR}(\text{x}_i) = \text{Norm}(\text{NA}(\text{x}_i, g)) \tag{4}$$

where $\text{Norm}(\cdot)$ is the layer normalization [47]. We assume that the input x_i is $S_i \times S_i \times d_i$. g is the spatial size of the homogeneous region. $\text{NA}(\cdot)$ is the operation of transforming x_i to several homogeneous regions, of which sizes $g \times g \times d_i$, and then a pixels' aggregation operation (i.e., mean filter) is performed on them and it produced an output of size $(S_i/g) \times (S_i/g) \times d_i$. The output is projected into SR(K) and SR(V). The attention score is calculated in the same way as ViT does [46], i.e., using all Q, K, and V as follows:

$$\text{Attention}(\text{Q}, \text{K}, \text{V}) = \text{softmax}\left(\frac{\text{QK}^{\text{T}}}{\sqrt{d_{\text{head}}}}\right)\text{V}. \tag{5}$$

To represent multiscale features, a reusable classification token is introduced to learn the long-range dependencies by

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including the feature tokens in each transformer encoder stage. After finishing the transformer block of the last stage, the classification token is fed to the linear layer, and then a softmax layer is used to calculate the probabilities for final classification. The final labeling of pixels is decided by the category with the largest probability value.

D. Implementation Details

Here, the Pavia University dataset, which consists of 610×340 pixels and 103 bands, is selected as an example to illustrate the implementation details of the proposed method. After PCA, patch feature extraction is used and obtains an input patch of size $15 \times 15 \times 30$. Then, a 3-D convolution layer with 32, $8 \times 8 \times 3$ cube kernels is performed on the input and generates a convolutional feature of size 32, $8 \times 8 \times 28$. The obtained feature is rearranged to one feature cube of size $8 \times 8 \times 896$ and fed to the proposed hierarchical transformer backbone network.

In the first stage of the transformer architecture, the feature cube after convolution is flattened to a feature map of size 64×896 and fed to a linear projection to generate embedded tokens of size 64×32. Then, the pixel tokens are concatenated with a reusable classification token T_0^{cls}, which has the same size as the embedded token for learning multiscale features. The obtained tokens along with a position embedding are passed through the proposed transformer block, and then the reusable token is upgraded as T_1^{cls} of size 1×32, separated from all the tokens, and prepared for the next transformer block. The resulting feature is reshaped to $8 \times 8 \times 32$ and used as output for the next stage.

From the second stage, the input is divided into 16 feature patches of size $2 \times 2 \times 32$ when the homogeneous region size is set to 2. The proposed NAE operation is performed on each patch and forms a set of new tokens of size 16×32, and the linear projection is used to form embedded tokens of size 16×64. Then, an additional fully connected layer is added to adjust the dimension of the classification token to match the pixel tokens, and the classification token T_1^{cls} of size 1×64 is concatenated with the pixel tokens. In the same way, we use the proposed transformer block and obtain T_2^{cls}. The remaining tokens are reshaped to a 3-D feature cube of size $4 \times 4 \times 64$ and used for the next stage.

In the last stage, using the feature cube and T_2^{cls} as input, we obtain the resultant feature of size $2 \times 2 \times 64$ and the classification tokens T_3^{cls} by carrying the same operations in Stage 2. Finally, a fully connected layer and a softmax layer are used on T_3^{cls} to form the final classification results.

IV. Experiments

A. Datasets Description

This section reports the classification experiments conducted on three public HSI datasets (i.e., Indian Pines, Pavia University, Houston 2013, and Houston 2018). The details of the aforementioned datasets are described below.

a) *Indian Pines Dataset*: The Indian Pines dataset was captured by an airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Northwestern Indiana in 1992. The dataset consists of 145×145 pixels and 200 spectral bands, with a spatial resolution of 20 m per pixel (24 water absorption channels were removed). As the scene was captured in June, some crops (such as corn and soybean) are still in the early stage of growth. The class name and the number of training and test samples are given in Table I.

b) *Pavia University Dataset*: The University of Pavia image was collected by the Reflective Optics System Imaging Spectrometer-3 (ROSIS-3) sensor over the University of Pavia and its surrounding in Pavia, Italy. The image comprises 610×340 pixels with a spatial resolution of 1.3 m per pixel and 115 bands. After removing 12 noisy bands, the experiments were conducted on the remaining 103 bands. A fixed number of samples for training and test were used as described in Table II.

c) *Houston 2013 Dataset*: This scene was acquired by the ITRES CASI-1500 sensor over the campus of the University of Houston and its neighboring rural areas, and it was used in the 2013 GRSS Data Fusion Contest. The data comprises 144 spectral bands in the 380–1050 nm region and 349×1905 pixels with a spatial resolution of 2.5 m. This image contains 15 classes,

TABLE I

Name and Number of Samples of Each Class on the Indian Pines Datasets With 10% Labeled Data

Houston 2018 dataset			
No.	Class Name	Train	Test
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture-mowed	48	435
6	Grass-trees	73	657
7	Grass-pasture	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	245	2210
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Buildings	39	347
16	Stone	6	84
	Total	1024	9225

TABLE II

Name and Number of Samples of Each Class on the Pavia University Datasets With 5% Labeled Data

Pavia University dataset			
No.	Class Name	Train	Test
1	Asphalt	332	6299
2	Meadows	932	17717
3	Gravel	105	1994
4	Trees	153	2911
5	Painted metal sheets	67	1278
6	Bare soil	251	4778
7	Bitumen	67	1263
8	Self-blocking bricks	184	3498
9	Shadows	47	900
	Total	2138	40638

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TABLE III

NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE HOUSTON 2013 DATASETS WITH 10% LABELED DATA

Houston 2013 dataset			
No.	Class Name	Train	Test
1	Healthy grass	125	1238
2	Stressed grass	125	1241
3	Synthetic grass	70	690
4	Trees	124	1231
5	Soil	124	1229
6	Water	33	321
7	Residential	127	1255
8	Commercial	124	1231
9	Road	125	1239
10	Highway	123	1214
11	Railway	123	1222
12	Parking Lot 1	123	1220
13	Parking Lot 2	47	464
14	Tennis Court	43	423
15	Running Track	66	653
	Total	1502	14871

TABLE IV

NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE HOUSTON 2018 DATASETS WITH 5% LABELED DATA

Houston 2018 dataset			
No.	Class Name	Train	Test
1	Healthy grass	595	9204
2	Stressed grass	1768	30734
3	Synthetic grass	36	648
4	Evergreen trees	685	12910
5	Deciduous trees	252	4769
6	Soil	226	4290
7	Water	14	252
8	Residential buildings	2001	37771
9	Non-residential buildings	11196	212556
10	Roads	2359	43507
11	Sidewalks	2060	31969
12	Crosswalks	78	1440
13	Major thoroughfares	2365	43983
14	Highways	494	9371
15	Railways	348	6589
16	Paved parking lots	576	10924
17	Unpaved parking lots	8	138
18	Cars	354	6193
19	Trains	268	5101
20	Stadium seats	342	6482
	Total	26025	478831

where the numbers of training and test samples are detailed in Table III.

d) *Houston 2018 Dataset* [48]: The data were acquired by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and its neighborhood on February 16, 2017. These data covering a 380–1050 nm spectral range with 48 bands at a 1-m ground sampling distance (GSD). The image consists of 601 × 2384 pixels. The ground truth of this image representing 20 urban land-cover/land-use classes is available. The numbers of sample for training and test are given in Table IV.

B. Experimental Setting

1) Quality Indexes: Three widely used quality indexes—overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ)—are calculated by averaging the results achieved after ten repeated experiments with various randomly selected training sets.

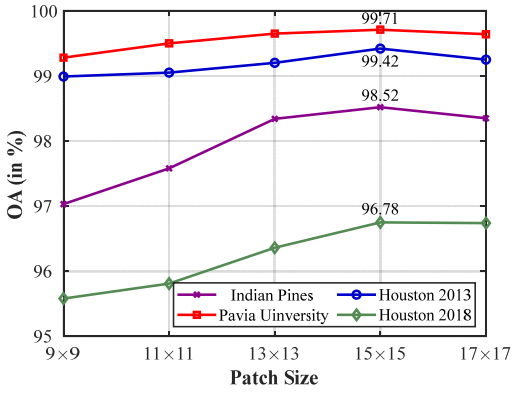


Fig. 3. Sensitivity analysis for the proposed method with different sizes of input patches.

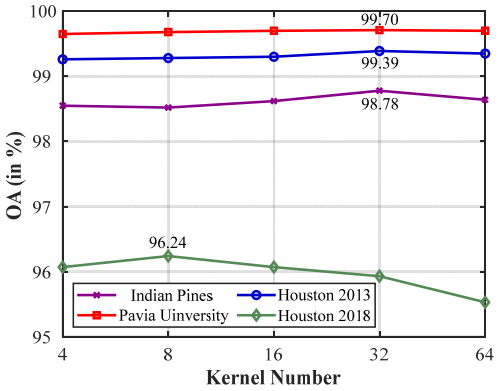


Fig. 4. Sensitivity analysis of the proposed method with different numbers of 3-D convolutional kernels.

2) Competitive Approaches: Three kinds of representative classification methods are adopted for comparison.

a) *Conventional Methods:* We compare with three standard methods: SVM, extended morphological profiles (EMP) [49], and superpixelwise PCA (SPCA) [50]. For the SVM, the libsvm toolbox is selected for the implementation of the classification task. The SVM is performed using the radial basis function (RBF) kernel. In the RBF, two hyperparameters σ and λ are optimally determined by fivefold cross validation on the training set in the range of $\sigma = [2^{-3}, 2^{-2}, \ldots, 2^4]$ and $\lambda = [10^{-2}, 2^{-1}, \ldots, 10^4]$, respectively. For the PCA used in the EMP, the number of principal components was set to three. The parameter settings were identical to those of the SVM above. For the SPCA, the number of superpixels is 100, 20, 600, and 700 on Indian Pines, Pavia University, Houston 2013, and Houston 2018, respectively.

b) *CNN Backbone Networks:* We use 2-D-CNN, 3-D-CNN, and spectral-spatial residual network (SSRN) [40] for comparison. For 2-D-CNN, three convolutional blocks, an average-pooling layer, and a softmax layer are used. Each convolutional block consists of a 2-D conventional layer, a batch normalization (BN) layer, a ReLU activation function, and a max-pooling layer. The patch size is 11 × 11. Moreover, the spatially and spectrally receptive fields in each 2-D convolutional layer are 3 × 3 × 32, 3 × 3 × 64, and 3 × 3 × 128 in size, respectively. The 3-D-CNN

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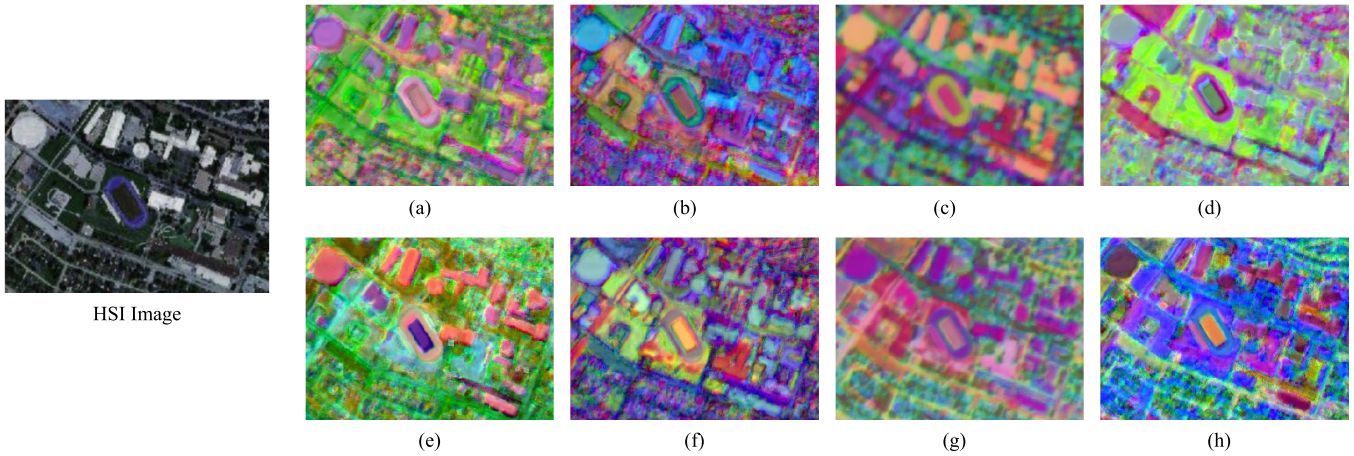


Fig. 5. Feature visualizations of selected heads by MHA and NAA. (a) MHA Head 1. (b) MHA Head 5. (c) MHA Head 32. (d) MHA Head 64. (e) NAA Head 1. (f) NAA Head 5. (g) NAA Head 32. (h) NAA Head 64.

architecture has three 3-D convolutional blocks and two linear layers. Similar to 2-D-CNN, the basic network block includes a 3-D conventional layer, a BN layer, a ReLU activation function, and a max-pooling layer. For SSRN, the patch size is 11 × 11 with 200 epochs for all the datasets.

c) *Transformer Architectures:* We compare the proposed method with SF, ViT, and spectral-spatial feature tokenization transformer (SSFTT) [35], which are the representatives of the transformer architecture. For the SF, cross-layer adaptive fusion modules are adopted in the transformer encoder stage, and four encoder blocks are used in the ViT-based network for HSI classification. For SSFTT, the network setting was as described in the following reference [35].

TABLE V

SENSITIVITY ANALYSIS OF THE PROPOSED METHOD AND COMPARISON OF DIFFERENT EMBEDDING DIMENSIONS ON THREE DATASETS

Data	Stage	OAs (in%) of Different Dimensions				
		32	64	128	256	512
Indian Pines	Stage 1	98.50	**98.68**	98.65	98.57	97.36
	Stage 2	98.70	**98.82**	98.87	98.60	97.41
	Stage 3	98.58	98.77	**99.00**	98.65	98.69
Pavia University	Stage 1	**99.43**	99.37	99.32	99.32	99.06
	Stage 2	99.30	**99.60**	99.57	99.47	99.05
	Stage 3	99.59	**99.72**	99.68	99.63	99.19
Houston 2013	Stage 1	99.04	**99.25**	99.22	99.19	97.30
	Stage 2	99.26	**99.32**	99.31	99.20	98.85
	Stage 3	99.13	99.28	**99.48**	99.37	98.55
Houston 2018	Stage 1	96.19	**96.27**	96.24	96.13	96.09
	Stage 2	96.31	96.37	96.40	**96.41**	96.37
	Stage 3	96.08	96.28	96.46	**96.70**	96.41

C. Model Analysis

1) Parameter Analysis: Several hyperparameters in the transformer encoder are set in advance. In the transformer architecture, the reduction ratio (i.e., g_i) is correlated with the size of input patch (i.e., S). Considering the depth and complexity of our model, S and g are set to 8 and 2, respectively, which are suitable for all the datasets. Therefore, the token number is 64, 16, and 4 for Stage 1, Stage 2, and Stage 3, respectively. In each transformer block, the number of heads (i.e., h_i) is recommend to set to 4, 8, 4, and 2 for the Indian Pines, Pavia University, Houston 2013, and Houston 2018 datasets, respectively. Besides, we found that the number of transformer encoders (i.e., L_i) has a little influence in our model, and L_i is set to 2 per stage for all the datasets, which is enabled for effectively capturing nonlocal information.

Furthermore, we also analyzed the impact of several critical parameters on the proposed method, i.e., the size of input patch W, the number of kernels in 3-D-CNN, and the embedding dimension d_i of each stage. Fig. 3 illustrates the classification performance of different sizes of input patches for the proposed method on the three considered datasets. The range of patch sizes is $\{9 \times 9, 11 \times 11, \ldots, 17 \times 17\}$. In all the tested datasets, the classification performance improves at first with the patch size enlarged and then decreased when reaching the best result. Therefore, a size of input patch of 15×15 is identified as adequate for all the datasets. Then, the experiments were conducted on the four datasets to analyze the impact of the number of 3-D kernels in the proposed method. The classification accuracies are given in Fig. 4. The results demonstrate that the OAs hold a stable and rising trend with the number of kernels. According to the best classification performance, the number of kernels is set to 32 for the Indian Pines, Pavia University, and Houston 2013 datasets, and 8 for the Houston 2018 dataset.

Besides, we investigated the sensitivity of the proposed method by comparing different embedding dimensions per stage. As shown in Table V, the experiments are conducted with four datasets, and the range of dimensions is set to $\{32, 64, \ldots, 512\}$. For the Indian Pines dataset, we first investigated the influence of the embedding dimension in Stage 1, which is 64 according to the optimal performance. Based on Stage 1 (i.e., $d_1 = 64$), the dimension of Stage 2 is analyzed and set to 64. Then, we tested the impact of the dimension in Stage 3 when Stages 1 and 2 each have an embedding dimension of 64. Therefore, the embedding dimension of all the stages is 64, 64, and 128, respectively. In this way, we determined the embedding dimension of all the stages on Pavia University (i.e., 32, 64, and 64), Houston 2013 (i.e., 64, 64, and 128), and Houston 2018 (i.e., 64, 256, and 256).

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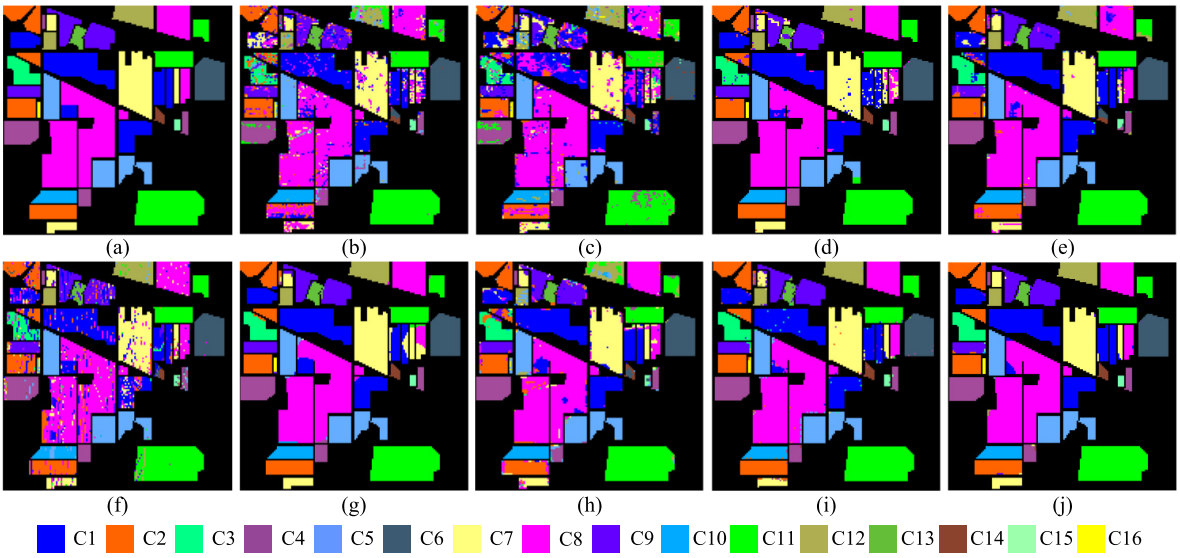


Fig. 6. Classification maps obtained by the compared methods on the Indian Pines dataset. (a) Reference map. (b) SVM. (c) EMP. (d) SPCA, (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) SF. (i) SSFTT. (j) LSFAT.

TABLE VI
ABLATION ANALYSIS FOR THE PROPOSED MODEL WITH A COMPARISON OF DIFFERENT EMBEDDING AND ATTENTION MODULES ON THE HOUSTON 2013 DATASET WITH 10% LABELED DATA

Layer	Embedding		Attention		Metric		
	ViT	NAE	ViT	NAA	OA(%)	AA(%)	$\kappa \times 100$
1-Layer	✓	×	✓	×	98.21	98.26	98.15
1-Layer	✓	×	×	✓	**99.32**	**99.39**	**99.27**
3-Layer	✓	×	✓	×	98.45	97.83	98.48
3-Layer	×	✓	✓	×	99.34	99.39	99.29
3-Layer	×	✓	×	✓	**99.69**	**99.70**	**99.67**

2) Ablation Study: For this part, we analyzed the performance improvement of the proposed NAE and NAA modules under different transformer structures. The ablation analysis experiment was conducted on the Houston 2013 dataset with 10.0% labeled data, and the classification performances of the proposed method are displayed on Table VI in terms of OA, AA, and Kappa. In detail, case 1 (i.e., ViT) consists of the spectral–spatial feature extraction module and a one-layer transformer structure without any specific module, yielding a relatively low classification result in all the cases. For case 2, a single-layer transformer structure with the NAA module obtained better classification result than case 1 (improved 1.11% in OA). Under the hierarchical transformer architecture, we found that the NAE module brings a performance improvement with regards to cases 3 and 4. By introducing the NAE and NAA modules into our hierarchical framework, the classification performance of case 5 is better than that reported for other cases, which demonstrates the effectiveness of the proposed NA-based modules.

Fig. 5 displays the visual maps of selected head features, extracted by MHA and NAA. We select the feature maps of head 1, 5, 32, and 64 from the transformer encoder to represent the whole encoding feature. As the figure shows, the NAA modules generate finer results in terms of structural information, preserving features better than MHA. It also demonstrates that our transformer architecture is capable of characterizing large-scale objects by means of the proposed homogeneous region aggregation strategy.

D. Prediction Results Evaluation

1) Random Sampling Analysis: In this part, we quantitatively analyzed the classification performance of the compared methods in terms of classification accuracies and visualization maps. The classification results of all the methods on the Indian Pines, Pavia University, and Houston 2013 datasets are given in Tables VII–X and Figs. 6–8. To be fair, the experiments were repeated ten times to obtain the average values of OA, AA, Kappa, and the accuracies for each class.

The first comparison experiment was conducted on the Indian Pines dataset with 10% of the reference data. Fig. 6 reports the obtained results. For SVM and EMP, significant noise exists in their visual maps [see Fig. 6 (b) and (e)], owing to the fact that manual features usually cannot work well with a large number of training samples. Moreover, 2-D-CNN, 3-D-CNN, and SF had noisy results. It can be found that SPCA, SSRN, and SSFTT yielded good classification performances in comparison to other methods, but our proposed method outperformed them. However, a few noisy spots remain around the edges among different objects [see Fig. 6 (h) and (i)]. In contrast, the proposed method is able to form a quite clean classification map when compared with other methods. Table VII gives the classification results of the comparison methods and the proposed method in terms of OA, AA, Kappa, and accuracies of each class. Among them, SSRN and SSFTT yielded better classification results than other studied methods, owing to their excellent contextual information modeling ability. It is clear that the proposed method achieved the highest classification value with respect to other studied approaches by integrating the properties of multiscale features and transformer architectures.

The second experiment was performed on the Pavia University dataset with 5% of the labeled data to demonstrate the classification performance of the prosed method.

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TABLE VII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods			CNN Backbone Networks			Transformer Architectures		
	SVM	EMP	SPCA	2D-CNN	3D-CNN	SSRN	SF	SSFTT	LSFAT
1	90.79	57.32	97.56	30.00	37.14	**100.0**	50.40	97.80	93.70
2	77.56	62.56	92.22	91.35	77.39	**97.80**	88.74	94.14	97.19
3	79.78	74.52	98.93	89.84	75.11	97.24	89.06	98.06	**99.43**
4	66.60	55.40	92.02	98.62	73.20	**100.0**	80.28	98.63	99.29
5	90.48	81.26	**99.31**	94.86	88.48	97.97	94.29	97.81	98.66
6	89.00	95.97	**100.0**	99.29	98.62	97.43	97.51	98.75	99.20
7	96.91	81.20	95.65	73.81	56.19	**100.0**	42.66	92.40	**100.0**
8	94.41	96.23	99.53	**100.0**	99.50	99.73	99.06	99.83	**100.0**
9	77.20	73.33	**100.0**	43.33	33.99	**100.0**	43.51	56.11	92.77
10	79.78	80.54	96.46	92.47	77.95	90.96	89.75	97.61	**98.45**
11	80.65	84.02	98.55	95.82	83.92	**99.45**	93.04	98.48	99.40
12	77.92	48.71	90.45	84.58	71.32	96.12	76.49	94.41	**98.33**
13	93.84	97.30	99.46	**100.0**	88.15	94.73	99.63	98.69	99.78
14	92.35	89.03	99.82	98.14	97.23	98.88	94.48	99.53	**99.97**
15	72.53	65.27	98.85	96.27	89.29	**100.0**	75.93	97.28	99.56
16	99.03	80.48	91.95	**99.29**	70.84	92.81	88.49	94.45	92.02
OA (%)	83.00	78.52	97.18	94.17	84.18	97.57	90.35	97.53	**98.90**
AA (%)	84.93	76.45	96.92	86.73	76.15	97.70	81.46	94.63	**98.00**
$\kappa \times 100$	80.54	75.39	96.78	93.35	81.94	97.24	89.06	97.18	**98.75**

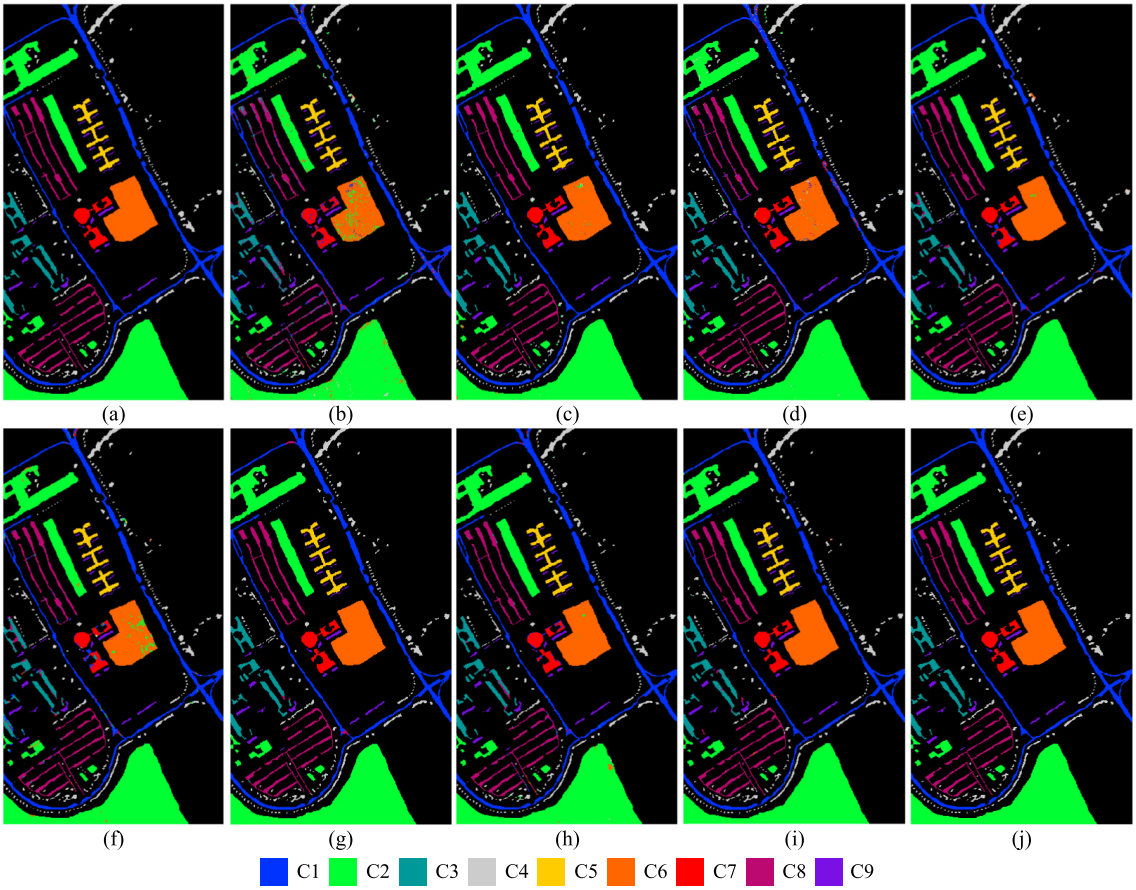


Fig. 7. Classification accuracies of the compared methods for the Pavia University dataset. (a) Reference map. (b) SVM. (c) EMP. (d) SPCA. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) SF. (i) SSFTT. (j) LSFAT.

Fig. 7 and Table VIII illustrate the classification maps and accuracies of the studied methods. As Fig. 7 shows, SVM, EMP, 3-D-CNN, and SF yielded different levels of noisy estimation in their visual maps by observing the brown region (i.e., bare soil class). SPCA, 2-D-CNN, and SSFTT performed better than the aforementioned methods with sufficient training data [see Fig. 7 (d)–(i)]. In contrast, the proposed method also yielded a fine visual map in terms of noisy estimation. Table VIII gives the classification results of the compared methods on the Pavia University dataset, where our method also obtained the best performance in comparison to the studied methods in terms of OA, AA, and Kappa.

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TABLE VIII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR PAVIA UNIVERSITY. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods			CNN Backbone Networks			Transformer Architectures		
	SVM	EMP	SPCA	2D-CNN	3D-CNN	SSRN	SF	SSFTT	LSFAT
1	93.64	99.14	95.11	98.87	96.23	99.29	95.39	98.58	**99.79**
2	95.63	99.64	99.60	99.95	99.06	99.16	99.72	**99.93**	99.86
3	84.02	**98.98**	98.50	94.01	83.19	91.02	89.84	96.73	98.15
4	97.00	97.79	94.85	98.71	97.63	**99.73**	97.23	97.88	99.33
5	99.41	99.66	98.98	99.41	99.39	**99.99**	99.95	97.16	99.89
6	92.58	98.63	97.24	99.57	94.12	99.12	92.01	**99.98**	99.93
7	86.44	99.33	94.70	91.39	80.26	98.17	90.91	99.82	**99.95**
8	85.25	98.62	93.34	98.90	96.58	97.90	94.97	98.78	**99.00**
9	99.98	99.82	99.56	99.68	99.38	**100.0**	99.11	96.90	98.80
OA (%)	93.54	99.19	97.52	98.98	96.38	98.66	96.79	99.17	**99.65**
AA (%)	92.66	99.07	96.87	97.83	93.99	98.27	95.46	98.42	**99.42**
$\kappa \times 100$	91.39	98.92	96.71	98.64	95.20	98.22	95.73	98.90	**99.53**

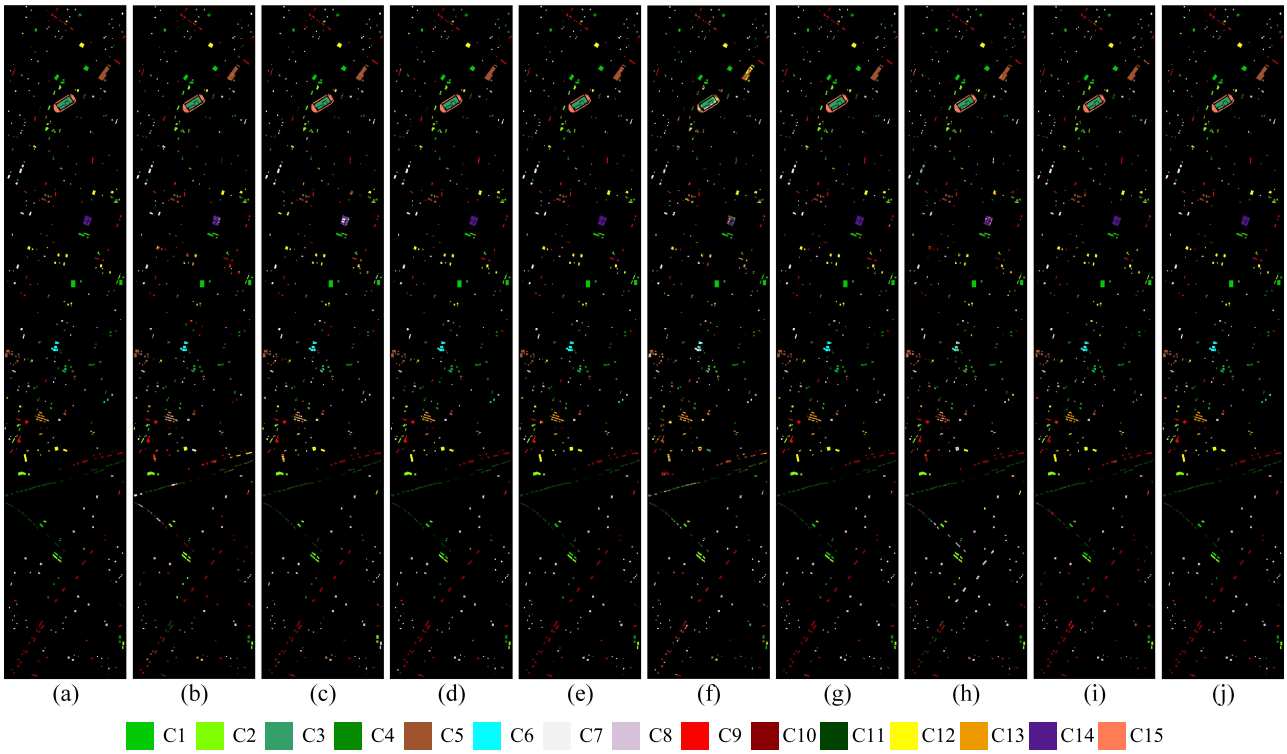


Fig. 8. Classification maps by the compared methods for the Houston 2013 dataset. (a) Reference map. (b) SVM. (c) EMP. (d) SPCA. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) SF. (i) SSFTT. (j) LSFAT.

The third comparison experiment was carried out on the Houston 2013 dataset with 10% of the training samples of the reference data. In Fig. 8, by observing the significant presence of large-scale buildings, e.g., tennis courts, parking lots, and running tracks, we can realize that the proposed method can efficiently preserve such large-scale structures. As Table IX shows, our method still achieved the best classification performance in comparison to other methods. Specifically, the accuracies of several classes reached 100%, i.e., soil, water, tennis court, and running track classes.

The last experiment was performed on the Houston 2018 dataset, and 5% of the labeled data were split form the reference data for training the classifiers. As shown in Fig. 9, the LSFAT achieves a high-quality classification map in comparison to other methods in terms of edge and texture. In Table X, we found that the deep-learning-based method cannot perform well in some challenge classes (e.g., water, crosswalks, and unpaved parking lots classes) with limited training samples. In contrast, the proposed method achieves better performance in comparison to them.

2) Nonoverlapped Sampling Analysis: The spectral–spatial classification methods benefit from the random sampling strategy because of the overlap between the training and test sets. However, the random sampling strategy is not flexible and difficult to implement in practical applications. A nonoverlapped sampling strategy [51] is adopted to form the training and test sets by following the number of Tables I–IV. Figs. 10–13 display the spatial distribution of the training and test sets. Then, the experiment was conducted on four datasets with the nonoverlapped training and test sets. In Table XI, it is concluded that the classification performance of all the methods dropped dramatically in comparison to random sampling.

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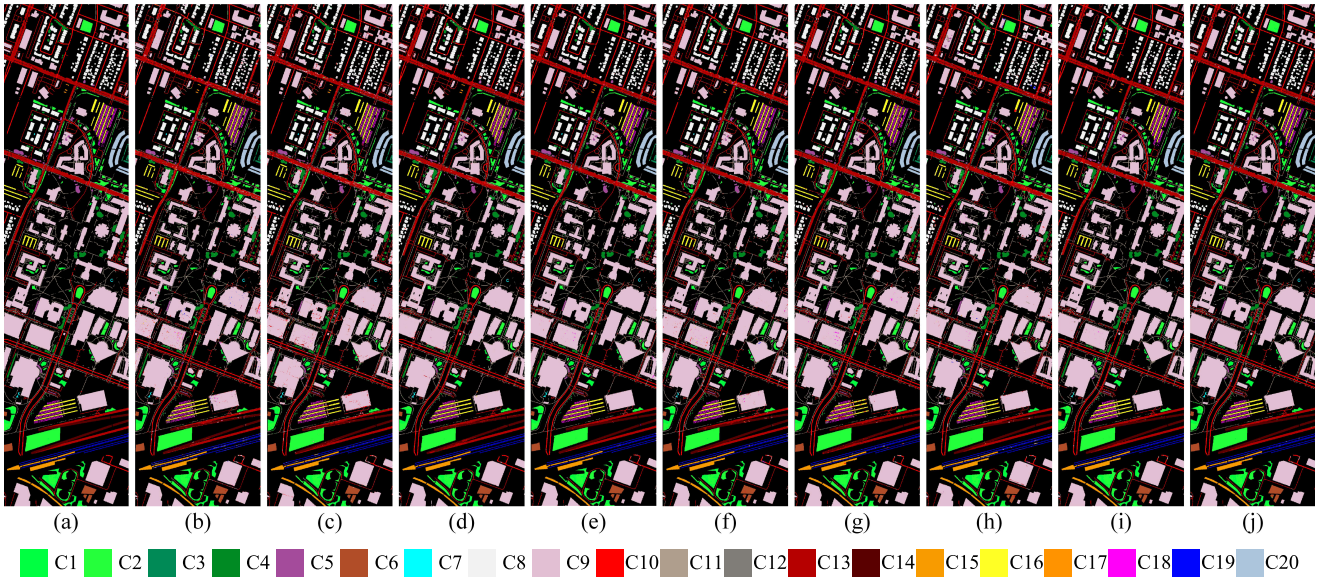


Fig. 9. Classification maps by the compared methods for the Houston 2013 dataset. (a) Reference map. (b) SVM. (c) EMP. (d) SPCA. (e) 2-D-CNN. (f) 3-D-CNN. (g) SSRN. (h) SF. (i) SSFTT. (j) LSFAT.

TABLE IX

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods			CNN Backbone Networks			Transformer Architectures		
	SVM	EMP	SPCA	2D-CNN	3D-CNN	SSRN	SF	SSFTT	LSFAT
1	96.93	99.56	98.37	97.70	94.02	**99.65**	98.27	98.45	99.00
2	97.85	99.01	97.31	99.90	94.46	99.81	98.88	99.52	**99.92**
3	99.49	**100.0**	**100.0**	**100.0**	99.86	**100.0**	99.44	99.39	99.98
4	98.93	**99.48**	91.40	97.37	96.85	98.57	95.20	98.76	99.02
5	96.86	99.79	99.01	99.89	95.08	**100.0**	99.57	99.78	**100.0**
6	99.86	98.25	95.24	95.56	91.61	**100.0**	84.03	99.72	**100.0**
7	92.11	98.52	87.51	93.20	95.15	**99.33**	91.26	95.64	98.37
8	87.51	96.07	93.25	91.49	83.70	98.12	90.83	97.39	**98.83**
9	85.34	95.75	92.28	94.89	88.78	92.95	90.66	98.15	**98.88**
10	92.54	98.42	97.45	98.83	95.42	95.89	96.46	99.70	**99.91**
11	90.95	96.83	97.82	99.15	95.79	98.31	93.23	99.11	**99.98**
12	87.84	97.00	96.59	98.09	88.00	96.14	94.95	99.49	**99.62**
13	86.77	76.47	73.77	96.66	97.54	97.90	60.54	96.12	**98.19**
14	97.90	98.60	**100.0**	**100.0**	99.44	**100.0**	96.88	99.58	**100.0**
15	99.13	99.71	99.53	**100.0**	93.84	**100.0**	98.00	99.07	**100.0**
OA (%)	93.42	97.56	94.98	97.35	93.22	98.10	94.05	98.63	**99.41**
AA (%)	94.00	96.90	94.63	97.52	93.97	98.45	92.60	98.66	**99.45**
$\kappa \times 100$	92.88	97.36	94.57	97.13	92.67	97.95	93.57	98.52	**99.36**

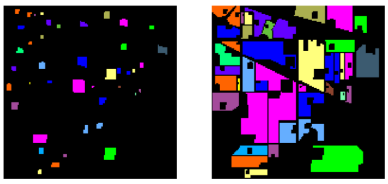


Fig. 10. Spatial distribution of non-overlapped training and test sets on the Indian Pines dataset. (a) Training set. (b) Test set.

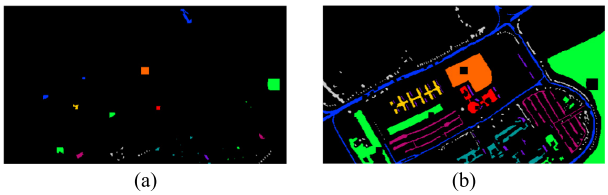


Fig. 11. Spatial distribution of non-overlapped training and test sets on the Pavia University dataset. (a) Training set. (b) Test set.

Our LSFAT outperforms the other comparison methods on most of the used datasets. SSRN achieves better classification results on the Indian Pines dataset. The reason is that the NA operation will remove the unique spectral characteristics of the training samples and make them similar when the samples are dense. In future, we aim to design a specific module for learning the sequential properties in the spectral dimension to overcome this problem.

3) Model Sensitivity Analysis: We randomly selected a varying proportion of training data from the reference data on the Indian Pines, Pavia University, Houston 2013, and Houston 2018 datasets, respectively. Fig. 14 illustrates the classification performance of the comparison methods and

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TABLE X

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2018 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods			CNN Backbone Networks			Transformer Architectures		
	SVM	EMP	SPCA	2D-CNN	3D-CNN	SSRN	SF	SSFTT	LSFAT
1	94.21	90.16	92.79	82.63	88.64	89.98	**97.43**	87.01	88.54
2	95.48	95.95	96.82	**96.95**	96.03	94.11	89.62	91.31	95.11
3	98.63	100.0	100.0	99.27	98.77	**100.0**	99.69	99.71	99.42
4	91.79	95.28	95.81	98.61	98.52	98.66	97.13	98.63	**99.04**
5	86.35	87.94	92.20	**94.13**	88.68	90.40	86.16	87.25	92.25
6	96.38	95.57	99.67	99.53	97.13	99.51	98.46	99.87	**100.0**
7	**100.0**	98.02	90.48	97.66	65.48	0.00	44.31	97.37	94.36
8	90.75	97.22	97.67	97.12	96.82	96.42	96.43	**98.93**	98.80
9	95.15	95.97	98.25	98.76	98.47	**99.04**	97.25	98.83	98.87
10	75.35	75.25	89.41	87.21	82.68	92.08	81.12	89.40	**93.44**
11	77.36	63.57	82.26	84.53	81.43	88.28	79.38	**89.86**	86.93
12	36.62	19.63	6.94	31.29	23.68	0.00	3.79	**41.77**	40.45
13	79.52	85.09	92.02	94.42	88.94	**97.29**	93.47	95.43	95.53
14	85.82	89.70	86.81	**98.28**	96.28	91.54	86.78	98.03	98.17
15	97.33	98.19	**99.76**	99.45	99.50	92.81	99.09	98.66	99.61
16	92.26	93.57	96.97	96.61	94.24	96.70	93.81	97.60	**98.17**
17	75.41	100.0	**100.0**	88.03	42.03	0.00	0.00	69.86	**100.0**
18	90.37	83.16	88.18	96.85	96.59	83.89	94.72	97.14	**98.17**
19	91.02	92.98	95.22	99.12	96.28	95.73	95.12	98.57	**99.25**
20	99.22	99.17	99.77	99.87	99.29	99.80	99.49	99.99	**100.0**
OA (%)	89.99	90.27	94.80	95.50	94.02	95.85	92.98	96.00	**96.57**
AA (%)	87.45	87.82	90.05	92.02	86.47	80.31	81.66	91.76	**93.80**
$\kappa \times 100$	86.94	87.30	93.22	94.14	92.20	94.60	90.88	94.81	**95.55**

TABLE XI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS WITH NONOVERLAPPED TRAINING AND TEST SETS IN TERMS OF OA, AA, AND κ FOR THE USED DATASETS. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Dataset	Metric	SVM	EMP	SPCA	2D-CNN	3D-CNN	SSRN	SF	SSFTT	LSFAT
Indian Pines	OA (%)	68.62	59.42	50.50	70.84	48.64	**80.77**	55.39	76.12	79.18
	AA (%)	63.82	54.77	48.00	62.90	44.05	**68.29**	43.98	65.04	67.49
	$\kappa \times 100$	64.54	53.88	46.37	66.45	41.05	**77.95**	49.15	73.05	76.12
Pavia University	OA (%)	75.92	81.51	56.59	82.55	77.51	83.99	70.08	80.25	**84.16**
	AA (%)	76.23	87.72	54.78	**80.09**	71.15	82.09	75.33	77.17	80.03
	$\kappa \times 100$	69.69	76.14	46.68	76.89	70.06	79.15	62.19	73.70	**78.80**
Houston 2013	OA (%)	61.63	62.45	42.28	65.84	51.07	63.58	64.60	65.49	**66.67**
	AA (%)	71.57	67.89	49.40	69.49	51.94	68.95	65.76	69.29	**70.90**
	$\kappa \times 100$	58.78	59.68	38.16	63.29	47.34	60.97	62.00	62.98	**64.18**
Houston 2018	OA (%)	75.23	57.59	66.62	83.89	80.23	83.34	80.62	83.74	**84.06**
	AA (%)	72.41	62.37	69.17	78.45	75.21	71.32	68.28	78.14	**80.45**
	$\kappa \times 100$	68.26	48.08	58.48	78.22	74.86	78.54	74.14	78.39	**78.73**

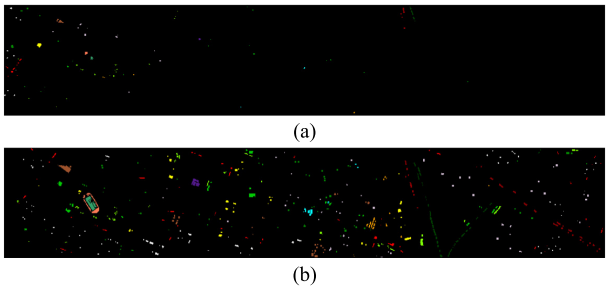


Fig. 12. Spatial distribution of non-overlapped training and test sets on the Houston 2013 dataset. (a) Training set. (b) Test set.

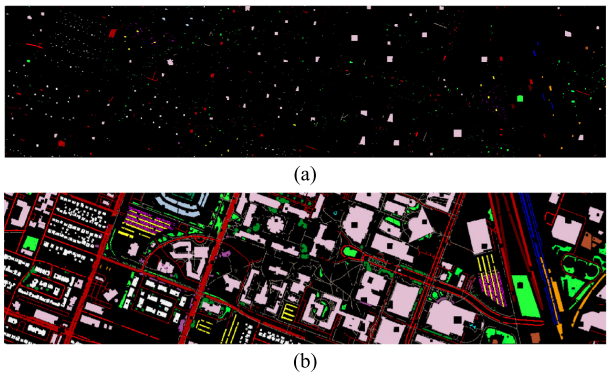


Fig. 13. Spatial distribution of non-overlapped training and test sets on the Houston 2018 dataset. (a) Training set. (b) Test set.

the proposed method with different numbers of training samples on three datasets. The percentages of the selected training samples on three datasets covers an interval of $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, $\{0.5\%, 1.0\%, \ldots, 5.0\%\}$, $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, and $\{0.5\%, 1.0\%, \ldots, 5.0\%\}$, respectively. As shown in Fig. 14(a), the classification performances gradually improve with the percentages of training samples. In addition, the OAs of the proposed method consistently outperform those achieved by other studied methods. The same conclusion can be extracted from the experiments with other three datasets; the proposed method achieves the most

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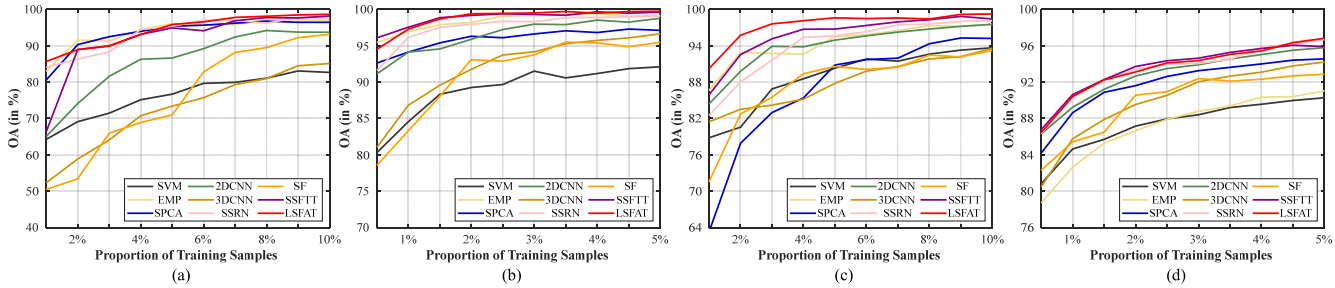


Fig. 14. Classification performance of the compared methods with different numbers of labeled data on three datasets. (a) Indian Pines. (b) Pavia University. (c) Houston 2013. (d) Houston 2018.

TABLE XII
COMPUTATION TIME (I.E., TRAINING TIME AND TEST TIME) IN SECONDS FOR THE DEEP-LEARNING-BASED METHODS

Dataset	Method	Training(s)	Test(s)
Indian Pines	2D-CNN	30.52	2.30
	3D-CNN	208.42	3.64
	SSRN	263.32	6.82
	SF	210.44	2.59
	SSFTT	19.00	0.57
	LSFAT	47.10	1.38
Pavia University	2D-CNN	75.10	7.75
	3D-CNN	276.91	11.87
	SSRN	95.22	3.68
	SF	570.76	6.84
	SSFTT	35.94	2.48
	LSFAT	88.53	5.65
Houston 2013	2D-CNN	85.49	2.37
	3D-CNN	101.98	5.45
	SSRN	80.61	1.59
	SF	466.13	2.72
	SSFTT	24.98	0.79
	LSFAT	66.49	2.03
Houston 2018	2D-CNN	1567.00	6.00
	3D-CNN	3549.99	129.60
	SSRN	1711.34	56.34
	SF	5188.14	101.81
	SSFTT	414.36	43.86
	LSFAT	674.05	65.90

satisfactory classification results with respect to other competitive methods. In conclusion, it is demonstrated that the proposed method offers a robust model is and able to achieve steady classification results.

4) Computational Cost: The proposed method has been implemented on the PyTorch platform using a desktop computer with i7-7800X CPU, 32-GB RAM, and an NVIDIA GTX 1080Ti 11-GB GPU. The Adam optimizer is used to optimize loss. The learning rate is 10^{-3}, and the batch size is 64 with 100 epochs on all the datasets. We investigated the computational cost of the compared methods. Table XII shows a comparison of the training and test times of several methods (i.e., 2-D-CNN, 3-D-CNN, SSRN, SF, and SSFTT) when compared with the proposed one on the Indian Pines, Pavia University, Houston 2013, and Houston 2018 datasets. The proposed method is faster than the other methods except for 2-D-CNN and SSFTT, because of the hierarchical transformer structure. It can be observed that SSFTT achieved the lowest training and test times, owing to its single-layer transformer structure. Besides, 2-D-CNN with three convolutional layers is close to the proposed method in terms of time consumption. 3-D-CNN and SF need a relatively long time for modeling the spectral information. SSRN used a deeper convolutional structure, which resulted in longer processing times. In summary, the proposed method is able to improve the classification performance significantly with an acceptable computational complexity.

V. CONCLUSION AND FUTURE WORK

The ViT has been shown to achieve excellent ability for modeling long-range dependencies. However, the transformer-based architectures cannot work well when attempting to learn the spatial contextual information of multiscale features. In this article, we introduce the concept of homogeneous regions into ViT and propose a novel LSFAT, which unifies multiscale feature extractions within a hierarchical transformer architecture. NAE and NAA modules and verify their effectiveness in comparison to ViT. Extensive classification experiments demonstrate that the proposed architecture provides good classification results and acceptable computational performance when compared with other state-of-the-art methods for HSI classification. Besides, the proposed NA modules are enable to supply a flexible shrinking representation for the input features while reducing the computational cost. However, our transformer architecture primarily consists of feature aggregation and self-attention calculation for exploiting long-range dependencies, and it ignores the specific design for capturing detailed discrepancies in spectral dimension.

In the future, we will further investigate the attention mechanism and improve the performance of the transformer-based architecture with self-supervised learning to overcome the limited number of training data. Besides, we desire to extend the applications of the proposed transformer architecture to active learning and self-supervised learning.

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